A Camera Zoom-based Paper-Pencil Cipher Encryption Scheme atop Merkle-Hellman Knapsack Cryptosystem

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Abstract

A Symmetric Key Encryption scheme using Camera Zooming is presented using a familiar Paper-pencil cipher. The Camera can have a magnification/scaling up to some integer. The encrypter and decrypter are two hardware systems that are assumed to have the capability to zoom a given image with text from the resolution of a single character to a page by applying an appropriate scaling factor and an appropriate polynomial time zoom algorithm. Using the symmetric key, the camera or a zooming algorithm implementation repeatedly zooms on different boxes in an image filled with cipher text and spurious redundant non-correlated pseudo-random data. Then, it decrypts the cipher using the Merkle-Hellman Knapsack Cryptosystem (MHKC) trapdoor algorithm or variants for efficient encryption/decryption. As will be shown, The MHKC algorithm's cryptanalysis vulnerability using Lattice-based LLL and other density attacks would not affect this scheme - as long as the key is private.

1 Introduction

Optical Character Recognition(OCR) is nearly reaching excellent accuracy in image and Handwriting recognition systems with the Advent of GenAI and other advances in Computer Vision. Today, a handwritten note can be easily parsed with OCR. Compelling evidence supports that Smart Camera-based hardware may soon take center stage in mainstream computing. Thus, camera algorithms, protocols, and security are actively researched. In this paper, the familiar Paper-pencil Stencil encryption scheme is analyzed for a Camera reading the Ciphertext rather than a human by using OCR, and a symmetric encryption scheme is proposed using fast computing of Merkle-Hellman Knapsack Cryptosystem Trapdoor(or variants using Chinese Reminder Theorem [4])

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as a Shared key to encrypt/decrypt. The shared key is exchanged in a public medium using industry-standard key exchange algorithms such as ECDH or RSA.

2 Paper-Pencil Cipher

A simple paper-pencil cipher is shown below with the hidden text

> Hello World

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$$
\begin{pmatrix}
A & B & K & \boxed{H & E} & F & C & S & O & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & \boxed{L & L} & R & R & B & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R \\
Q & \boxed{W} & E & R & T & y & C & S & \boxed{O} & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & a & b & c & O & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & S & R & R & \boxed{R} & A & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & S & R & R & \boxed{L} & \boxed{D} & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R \\
A & B & K & H & E & F & C & S & O & S & R & R & R & A & B & K & H & E & F & C & S & O & S & R & R & R
\end{pmatrix}
$$

With the camera's zoom capabilities, we could repeatedly zoom individually at boxes of varied lengths to figure out hidden text. This also obfuscates the plaintext with random data. If the entire matrix had been encrypted, the ciphertext first had to be located, and then individual boxes have to be decrypted to get the plaintext. In this paper, we look at the encrypted text as an image/matrix in the Camera context, but nothing precludes the entire Matrix from being treated as a stream of bytes/bits, and the positions of the Stencils set then become indices set into the stream.

3 Definitions

Let Stencil Set S correspond to the exhaustive list of stencils of all shapes of the red boxes shown. E.g., the text "RLD" is L-shaped.
We could also have other arbitrary shapes, such as disconnected ones containing

the ciphertext.
Thus, Stencil set S is the set of all possible shapes to use. The stencil set S is public.
Let M be the message. and $l = |M|$ denotes the length of a message.
Let R_l be a random partition of a random number $l_{max} \geq l$. i.e.

$$R_l = \{x_i \in Z^+ | \sum_{i=0}^{n} x_i = l_{max}\}$$

Let $S_{R_l} \subset S$ be a subset of shapes S s.t there is a bijection $g : R_l \to S_{R_l}$.
The requirement is that all the encrypted text of M fits within the Stencil sets in S_{R_l}
An example of an element of S_{R_1} for the picture above "RLD" Stencil set is

$$s_{rld} = \{(13, 12), (13, 13), (14, 13)\}$$

The first coordinate starts the Stencil assuming the fourth quadrant of the x-y plane with positive values for x, y. Here $s_{rld} \in S_{R_l}$. The encrypted text is split among the stencils and permuted, attaining enough permutation to be indistinguishable from surrounding random letters.
Let $\sigma(M)$ be the permutation of M post deciphering. we permute s.t $\sigma^g(M) = M$.
The Secret key is the set $\{S_{R_l}, g, Private_{mhkc}, \sigma, g\}$ The underlying encryption algorithm is Public key Cryptosystem with keys $\{Public_{mhkc}, Private_{mhkc}\}$ of Merkle-Hellman Knapsack cryptosystem using MHKC trapdoor [3]

4 Encryption/Decryption

The encryption/decryption follows MHKC super-sequence-based knapsack cryptosystem as per [3] Or other variants such as [4] for the whole matrix with pre-filled permuted message $\sigma(M)$ in a chosen order at the appropriate stencil set. If a default order is not specified, then the start of the text and the stencil square with the start of the message are identified/included as part of the secret key. By default, the camera is assumed to first zoom left-to-right top to bottom order for decryption. If a random zooming order on the stencils is enforced, e.g., for the Matrix in the picture, stencil with W occurs before O, then there is an extra step of re-permuting the obtained text to glean the meaning of the full message in polynomial time. In such a scenario, the Secret key need not contain the permutation information.

5 Cryptanalysis

The Lenstra et al.'s LLL-lattice basis reduction [2] cryptanalysis for polynomial low-density attacks is still applicable (This is a 2-D Lattice), but security parameter IND-CPA(Indistinguishability, Chosen Plain text) is reconciled with the random plaintext and the permutation of the actual plaintext in the deciphered matrix of cipher as shown in the above matrix.

6 Brief Strength Analysis

The strength of the cryptosystem lies in the randomness of the Stencil set S_{R_l} selected and the selection of a random partition R_l. And the indistinguishable structure that it offers. We focus on the probability of reaching the "actual" plaintext hidden within the matrix, given that the LLL-Lattice reduced-basis polynomial time reduction algorithm has been successfully run to decipher the Matrix (for MHKC or a Variant). This is a two-dimensional Lattice where even Gauss's basis reduction could be run in Polynomial Time to determine a reduced basis and, hence, a polynomial-time algorithm to arrive at the matrix.

The approximate probability of uniformly (uniform distribution) selecting a partition of a number $n = l_{max}$ is $P_1 = \frac{1}{P(n)}$ Where $P(n)$ is the classic mathematical partition function. Thus, approximately

$$P_1 = \frac{4n\sqrt{3}}{e^{\pi\sqrt{\frac{2n}{3}})}}$$

Next, the probability of selecting a stencil can be calculated as follows.
The number of elements of the partition so chosen, i.e., $k = |R_l| = |S_{R_l}|$, on average, is based on the probability distribution of the Message Space. The number of ways of selecting the stencil set is $\binom{|S|}{k}$; this is also the number of possible maps g. The probability of selecting a stencil set S_{R_l} is $P_2 = \frac{1}{\binom{|S|}{k}}$.
Thus the overall probability of correctly getting the Stencil set containing the Plain text in the indistinguishable matrix is just

$$P = P_1 P_2 = \frac{4n\sqrt{3}}{e^{\pi\sqrt{\frac{2n}{3}})}\binom{|S|}{k}}$$

as these are independent events. This means that by carefully choosing l_{max} and the stencil set size $|S|$ and with lower k, we can decrease the probability as low as possible, irrespective of the encryption scheme. In this case, the MHKC cryptosystem is chosen to provide a fast encryption/decryption scheme. For Example, A message of length 40 has $P_1 = \frac{1}{37338} = 0.00002...$ which is a very low probability.

7 Extensions and Complexity

In this scenario, the Matrix is assumed to be specific to a plaintext M. We can also have a predetermined matrix with the composition of various plain text, where a set of plain texts share the same random matrix, or we can have a standard alphabetical/ASCII matrix. Similarly, the stencil sets can have "random"

boxes as elements rather than "connected" boxes. The idea is to accommodate a part of the message corresponding to the value of the selected partition in the same stencil set. Again. Nothing precludes not honoring that requirement. This is only for efficiency.
If there is no Matrix for each message, rather a standard ASCII Matrix is referred to by encrypting and decrypting systems, like an agreed reference Matrix containing all possible messages from message space of M, then the time complexity is the Complexity of the key exchange for Stencil set exchanges, as there is no encryption of the plaintext at all.
In all cases, The worst-case complexity is Polynomial in the size of the Matrix with no added random text. To randomly generate a partition of n, the time complexity is $O(n^{\frac{3}{4}})$ using Fristedt Algorithm [1].
To randomly generate a Stencil shape set, we could use Vitter's algorithm or variant [5] or [6] in $O(n)$ time.
Hence, overall, a worst-case Polynomial asymptotic $O(n^{\frac{3}{4}})$ is achievable, assuming MHKC with multiple iterations can be run at high speeds.

8 Conclusion

Modern camera-based AI agent communication with vision helps process data asymmetrically using Cameras, Network protocols, and Human Collaboration. In such a Hybrid environment, seamless, secure transactions across domains are becoming a necessity rather than a theoretical interest while retaining existing Encryption schemes.

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